Akoizo Original Practice - Form 2 - Answer key

Division B | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Oxytocin-driven labor is an example of:

Unit: Homeostasis, Feedback, and Gradients.
Worked-case data: A variable begins 8 units above its reference. Each step corrects 25% of the remaining deviation. Predict the error after two steps.

A. Positive feedback

B. Feedforward only

C. No feedback

D. Negative feedback

Correct: A

2. (2 points) Type II pneumocytes produce:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. HCl

B. Antibodies

C. Mucus

D. Surfactant

Correct: D

3. (2 points) Exercising muscle unloads more O2 because of:

A. High pH

B. Left shift from alkalosis

C. Right shift from acid, CO2, heat

D. Low 2,3-BPG

Correct: C

4. (2 points) Apical cavitary disease with caseating granulomas suggests:

Unit: Infection and Restriction: Pneumonia, TB, Fibrosis.
Worked-case data: Case A has a focal air-space opacity and ongoing blood flow through poorly ventilated tissue. Case B has diffuse interstitial thickening and reduced total lung capacity.

A. Asthma

B. TB

C. Emphysema

D. Pneumothorax

Correct: B

5. (2 points) B12 is absorbed in the:

A. Jejunum

B. Terminal ileum

C. Colon

D. Duodenum

Correct: B

6. (2 points) Lactose intolerance is caused by deficiency of:

Unit: Lactose Intolerance, Obesity, and Exercise Metabolism.
Worked-case data: A teaching model supplies 24 g lactose during transit. In one setting, 18 g can be hydrolyzed; in another, only 6 g can be hydrolyzed.

A. Maltase

B. Pepsin

C. Sucrase

D. Lactase

Correct: D

7. (2 points) Hard fixed nontender supraclavicular node most suggests:

Unit: Spleen, Nodes, and Barriers in Defense.
Worked-case data: One case involves particles arriving from a skin wound in lymph. Another involves microbes circulating in blood. A third involves intestinal luminal exposure.
Separate investigation observations:
Input: The organ receives and filters blood.
Architecture: White pulp supports immune responses; red pulp processes erythrocytes.
Function: The case emphasizes defense against blood-borne material.

A. Vaccine reaction

B. Viral URI

C. Malignancy

D. Allergy

Correct: C

8. (2 points) CD8 T cells recognize antigen on:

A. MHC II only

B. MHC I

C. Free antibody

D. Complement

Correct: B

9. (2 points) Heat-worsened blurred vision with INO suggests:

Unit: Autoimmunity, MS, RA, and Systemic Allergy.
Worked-case data: Case A describes central myelin injury. Case B describes destructive synovial inflammation. Case C describes rapid systemic effects after re-exposure to an external antigen.
Separate investigation observations:
Timing: Effects develop rapidly after re-exposure.
Trigger: The trigger is an external antigen encountered previously.
Mechanism: Mast-cell mediators alter vascular permeability and airway smooth muscle.

A. MS

B. Gout

C. Celiac

D. RA

Correct: A

10. (2 points) High blood calcium triggers which hormonal response?

Unit: Homeostasis, Feedback, and Gradients.
Worked-case data: A variable begins 8 units above its reference. Each step corrects 25% of the remaining deviation. Predict the error after two steps.

A. More ADH

B. More aldosterone

C. More PTH

D. More calcitonin, less PTH

Correct: D

11. (2 points) Normal pleural pressure during quiet breathing is:

Unit: Lungs, Alveoli, Pleura, and Breathing Muscles.
Worked-case data: At one instant, alveolar pressure is 0 cm H_2O relative to atmosphere and pleural pressure is -7 cm H_2O. At another, alveolar pressure is -1 and pleural pressure is -6.

A. Equal to alveolar

B. Positive 10 mmHg

C. Atmospheric

D. Negative around -4 mmHg

Correct: D

12. (2 points) Most CO2 is carried as:

A. Dissolved gas

B. Carbamino only

C. Bicarbonate

D. Myoglobin

Correct: C

13. (2 points) Basilar honeycombing with Velcro crackles suggests:

Unit: Infection and Restriction: Pneumonia, TB, Fibrosis.
Worked-case data: Case A has a focal air-space opacity and ongoing blood flow through poorly ventilated tissue. Case B has diffuse interstitial thickening and reduced total lung capacity.

A. IPF/UIP fibrosis

B. Asthma

C. Effusion

D. COPD

Correct: A

14. (2 points) Celiac disease labs and biopsy show:

Unit: Small Intestine, Liver, Pancreas, and Absorption.
Worked-case data: A case supplies normal pancreatic enzyme production but greatly reduced bile delivery to the small intestine. Predict the most directly affected step.

A. High gastrin only

B. No antibodies

C. Anti-tTG + villous atrophy

D. Low lipase + normal villi

Correct: C

15. (2 points) Best test for lactose malabsorption:

Unit: Lactose Intolerance, Obesity, and Exercise Metabolism.
Worked-case data: A teaching model supplies 24 g lactose during transit. In one setting, 18 g can be hydrolyzed; in another, only 6 g can be hydrolyzed.

A. Colonoscopy

B. Anti-tTG

C. Hydrogen breath test

D. Lipase

Correct: C

16. (2 points) Mono with splenomegaly should avoid:

Unit: Spleen, Nodes, and Barriers in Defense.
Worked-case data: One case involves particles arriving from a skin wound in lymph. Another involves microbes circulating in blood. A third involves intestinal luminal exposure.
Separate investigation observations:
Input: The organ receives and filters blood.
Architecture: White pulp supports immune responses; red pulp processes erythrocytes.
Function: The case emphasizes defense against blood-borne material.

A. Rest

B. Contact sports

C. Hydration

D. Follow-up

Correct: B

17. (2 points) Placental antibody protection is mainly:

Unit: Adaptive Immunity, Antibodies, and Allergy.
Worked-case data: An experiment shows a delayed small antibody rise after first exposure and an earlier larger rise after a later exposure to the same antigen.

A. IgG

B. IgE

C. IgD

D. IgM

Correct: A

18. (2 points) Most specific RA antibody:

Unit: Autoimmunity, MS, RA, and Systemic Allergy.
Worked-case data: Case A describes central myelin injury. Case B describes destructive synovial inflammation. Case C describes rapid systemic effects after re-exposure to an external antigen.
Separate investigation observations:
Timing: Effects develop rapidly after re-exposure.
Trigger: The trigger is an external antigen encountered previously.
Mechanism: Mast-cell mediators alter vascular permeability and airway smooth muscle.

A. ASO

B. Anti-tTG

C. ANA

D. Anti-CCP

Correct: D